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A multi-source transfer learning model based on LSTM and domain adaptation for building energy prediction

Huiming Lu [a], Jiazheng Wu [a], Yingjun Ruan [a,*], Fanyue Qian [a], Hua Meng [a], Yuan Gao [b], Tingting Xu [a]

[a] College of Mechanical and Energy Engineering, Tongji University, Shanghai 200092, China
[b] Department of Architecture, Graduate School of Engineering, The University of Tokyo, Japan

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ABSTRACT

Transfer learning can use the knowledge learned from the operating data of other buildings to facilitate the energy prediction of a target building. However, most of the current research focuses on the transfer of a single source building of the same type or from the same region. A single source domain produces domain shift due to the difficulty of aligning the distribution with the target domain. To address this problem, this paper proposes a novel multi-source transfer learning energy prediction model based on long short-term memory (LSTM) and multi-kernel maximum mean difference (MK-MMD) domain adaptation. This model was used for the short-term energy prediction of different types of buildings lacking historical data. In addition, dynamic time warping (DTW) was used to select the source domain. Multiple multi-source models and corresponding single-source models were compared on a collection of buildings in the Higashida area of Fukuoka Prefecture, Japan. On the experimental datasets, the results showed that DTW relatively accurately measured the similarity between building energy consumption datasets. Compared with the results of the single-source transfer learning models, the multi-source transfer learning models achieved better average prediction performance, and their mean absolute percentage error (MAPE) improved the prediction accuracy by 6.88–15.37%.

1. Introduction

1.1. Background

According to statistics, building energy consumption accounts for 40% of global energy consumption [1]. The *China Building Energy Consumption Research Report 2020* [2] pointed out that in 2018, the national building energy consumption during the operation phase was 1 billion tons of coal equivalent (TCE), accounting for 21.7% of the total energy consumption in the country. It can be seen that the demand for energy consumption in buildings is relatively high and has great energy-saving potential. With the rapid development of urban informatization and intelligent construction, the creation of advanced building energy management systems is of great significance for improving the building energy utilization efficiency. The efficient and accurate forecasting of building energy consumption plays a crucial role in the operation and optimization of building energy management systems.

1.2. Energy prediction model

Traditional energy prediction methods include physical models based on thermodynamic mechanisms [3,4] and statistical models based on mathematical statistics [5,6]. The physical models consider the physical properties of building materials and the operating parameters of the building and conduct dynamic simulation of the building according to the heat balance equation, which has clear physical meaning. Numerous types of energy prediction software have been developed and implemented, such as EnergyPlus, TRNSYS, DOE-2 and DeST. However, building information is usually difficult to obtain, it is difficult for the models to adapt to changes in the environment, and the models take a long time to build, which limits the practical applications of this method. Statistical models, such as the differential autoregressive moving average model (ARIMA) [6], are usually constructed based on the combination of linear functions and make predictions based on the correlation of time series themselves. Nevertheless, they are limited by the large number of parameters to be determined and the requirement of

* Corresponding author.
E-mail address: ruanyj@tongji.edu.cn (Y. Ruan).

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stable and low-dimensional data.

With the continuous improvement of the dimension of building energy consumption data in recent years, deep learning models with the ability to process high-dimensional nonlinear data, such as long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN), can learn the mapping relationship between data from historical building energy consumption and meteorological data. These models show excellent performance in building energy prediction, which has attracted significant academic attention [7–11].

In traditional deep learning methods, to ensure the reliability and accuracy of training, there are two basic assumptions: that the training data and test data satisfy the same probability distribution and that there are enough samples to train the model [12]. However, in many practical applications, the data distribution of time series such as energy consumption data may change over time. In addition, new buildings or old buildings that need energy-saving renovation have little historical data, which prevents the model from achieving high accuracy, so the above assumptions may not be satisfied in some cases.

1.3. Transfer learning in energy prediction

Transfer learning enables energy prediction across buildings when data are insufficient [13–16]. Transfer learning relaxes the two basic assumptions in traditional deep learning; that is, training and testing data can have different distributions, and tasks can be done well even with a small amount of data [17]. In the field of energy prediction, transfer learning uses the knowledge learned from historical data of similar buildings to facilitate the energy prediction of target buildings so that the target buildings can obtain more accurate prediction data even when the amount of available data is small. As a framework, transfer learning combined with deep learning algorithms is increasingly used in the field of building energy prediction.

For instance, Fan et al. [18] proposed a deep learning model of a convolutional neural network (CNN) and LSTM, combined with the transfer learning method for pre-training and fine-tuning, to use the data of the past seven days to predict the building energy consumption one day in the future. This study simulated two scenarios of old buildings and new buildings, and the knowledge extracted from the pre-trained model was used for feature extraction and weight initialization, respectively. The results verified the importance of transfer learning for short-term load forecasting. Gao et al. [19] built two source models, seq2seq and a two-dimensional CNN with an attention mechanism, and fine-tuned the pre-trained models using one month's data of the target building. As a result, the prediction accuracy was close to that using the two-year historical data of the target building itself. Fan et al. [20] quantified the potential of algorithm-centric and data-centric transfer learning strategies to improve building energy forecasting using actual measurement data from 600 buildings. Zeng et al. [21] proposed a weighted instance-based holiday peak electricity load prediction framework, which could appropriately enrich a sparse holiday dataset by the weighted transfer of historical holiday data from a source city to a target city. Fang et al. [22] utilized adversarial learning of a domain-adversarial neural network (DANN) and LSTM network to extract common features of source and target buildings. Zhou et al. [23] proposed a short-term energy prediction framework based on bidirectional generative adversarial networks (BiGAN) and fine-tuning, employing BiGAN to implement dataset augmentation. The results showed that the proposed short-term energy prediction framework significantly improved the prediction performance.

1.4. Methods and applications of transfer learning

Parameter sharing (also known as parameter freezing) is the most widely used transfer learning method in building energy prediction, including two steps: pre-training and fine-tuning [19]. First, the source model is obtained by pre-training the neural network with more source domain data; then, most of the layer parameters of the source model are frozen; and finally, the last few layers are fine-tuned with less target domain data to generate the target network. Parameter sharing is practical, simple, and feasible. Due to the use of pre-trained models, the network does not need to be trained from scratch for new tasks, which reduces the time cost of model training. However, when the data distributions of the source domain and the target domain differ, the parameter sharing alone usually does not achieve a good transfer effect.

Deep domain adaptation (DDA) is another transfer learning method applied in the field of building energy prediction. DDA, a domain adaptation technique combined with deep learning algorithms, solves the problem of domain shift by aligning the data distributions of the source and target and extracting common features between the source and target domains. DDA includes three methods:

(1) Discrepancy-based domain adaptation: First proposed by Tzeng et al. [24], an adaptation layer and a discrepancy loss are added to the deep neural network to map source and target domains with different distributions to the same feature space. The difference is measured by the maximum mean difference (MMD, described in Section 3.3.2) by optimizing the network loss so that the latent features of the source and target domains are as close as possible in this space.
(2) Adversarial-based domain adaptation: Ganin et al. [25] introduced the adversarial idea into neural networks for the first time and proposed the DANN. The DANN contains a feature extractor and a domain classifier. The goal of the feature extractor is to extract the common latent features of the source and target domains, while that of the domain classifier is to distinguish the features from the source or target domains. Finally, the DANN extracts the common features of the source and target domains through the game.
(3) Reconstruction-based domain adaptation [26]: This method reconstructs the source and target domain data based on AE. Currently, reconstruction-based domain adaptation has few applications in the field of building energy prediction.

1.5. Research gap and our contribution

Although the deep transfer learning model can effectively solve most of the building energy prediction problems related to poor data, some research gaps remain:

(1) In terms of applications, most of the current research focuses on the transfer of a single source building of the same type or from the same region [19,27], or multiple source buildings are simply merged into a single source domain and migrated to the target building of the same type and from the same region [28]. Very few studies have focused on the transfer of multiple source buildings. Practical applications often involve multiple source buildings, and multi-source transfer learning can usually reduce domain shift and achieve better prediction results than single-source transfer learning [29].

For instance, Fang et al. [30] proposed a multi-source ensemble transfer learning framework (Multi-LSTM-DANN) that combines the LSTM-DANN and similarity measure, where the LSTM-DANN extracts common temporal features of source and target building data. The framework utilizes multi-source building data to improve the prediction performance of target building energy consumption. However, the study still focused on the transfer of the same type of buildings (office buildings).

(2) In terms of method, most of the current research focuses on parameter sharing (pre-training and fine-tuning) in transfer learning methods. However, fine-tuning usually cannot achieve satisfactory prediction performance when the probability distributions of source and target domains differ. DDA can solve this problem well. However, existing studies have focused on adversarial domain adaptation [23,30]. Discrepancy-based and reconstruction-based domain adaptation studies

are relatively rare.

To address these research gaps, this paper makes the following contributions:

(1) A novel multi-source transfer learning energy prediction model is proposed, which is transferred between different types of source and target buildings. The aim is to verify that source buildings can improve prediction accuracy when they are sufficiently similar to the target buildings, even if the type of building and target buildings are different. Our model is suitable for multi-source energy prediction of buildings with little historical data.
(2) Our proposed model integrates multiple subnetworks for multiple source domains, parameter sharing and MK-MMD domain adaptation (discrepancy-based domain adaptation) methods are used to extract the common temporal features of each pair of source–target buildings separately. A comparison of this model with the single-source transfer learning model proves the superiority of the multi-source transfer learning model, and supplementary experiments verify the versatility and stability of the multi-source model.

In addition, to address the selection of multiple source domains in transfer learning, we use dynamic time warping (DTW) as a similarity measure between source and target domains. By fitting the correlation between DTW values and the mean absolute percentage error (MAPE) reduction rate of the single-source model, it is verified that DTW can relatively accurately measure the similarity of building energy consumption distributions.

The paper is organized as follows: Section 2 provides the theoretical background of this paper. Section 3 elaborates on the research outline, model framework, methodologies used, and evaluation metrics. Section 4 verifies the DTW effect using a single-source transfer learning model on different types of building sets and presents the experimental results of multiple sets of multi-source transfer learning energy prediction models. Section 5 presents our conclusions.

2. Theory of multi-source transfer learning

Transfer learning aims to leverage knowledge from a related domain (called the source domain) to improve learning performance in the task domain (called the target domain) or reduce the amount of labeled data required for the task domain [17]. Transfer learning is of great significance for the practical application of advanced deep learning algorithms in real-world scenarios. However, the results produced by transfer learning are not always satisfactory [31,32]. If the similarity between the source and target domains is relatively low, there is domain shift [33], at which time the target learner will be negatively affected by the transferred knowledge, a phenomenon called negative transfer [32]. To solve this problem, many scholars began to study more realistic multi-source transfer learning [29,34–35]. Since multiple source domains have complementary effects on the target domain, multi-source transfer learning can usually reduce negative transfer and achieve better prediction results than single-source transfer learning. Before defining multi-source transfer learning, a few related definitions are introduced.

Definition 1 (Domain): A domain $\mathscr{D}$ is denoted by $\mathscr{D} = \{\mathscr{X}, P(X)\}$, where $\mathscr{X}$ represents the feature space, $P(X)$ represents the marginal distribution, and X denotes an instance set, which is defined as $X = \{\mathrm{x}|\mathrm{x}_i \in \mathscr{X}, i = 1, \cdots, n\}$.

Definition 2 (Task): A task $\mathscr{T}$ is denoted by $\mathscr{T} = \{\mathscr{Y}, f\}$, where $\mathscr{Y}$ represents the label space and f represents an objective predictive function. f represents the mapping relationship between $\mathscr{X}$ and $\mathscr{Y}$, $f(\mathrm{x}_j) = \{P(y_k|\mathrm{x}_j)|y_k \in \mathscr{Y}, k = 1, \cdots, |\mathscr{Y}|\}$. In this paper, the label space includes the energy consumption of the building to be predicted.

Definition 3 (Multi-source transfer learning): Given some source domains and tasks (i.e., $\{(\mathscr{D}_{S_i}, \mathscr{T}_{S_i})|i = 1, \cdots, m^S\}$) and some target domains and tasks (i.e., $\{(\mathscr{D}_{T_j}, \mathscr{T}_{T_j})|j = 1, \cdots, m^T\}$), multi-source transfer learning utilizes the knowledge implied in the source domains to improve the performance of the learned objective predictive functions $f^{T_j}(j = 1, \cdots, m^T)$ on the target domains. Note that there is only one target domain and task for multi-source transfer learning in this study; therefore, $m^T = 1$.

At present, there are relatively few studies on multi-source transfer learning, and two application strategies are mainly used: the first strategy is model ensemble, which aims to use TradaBoost [36] and its improved algorithm to combine the obtained weak classifiers into strong classifiers by iteratively updating the weights of instances. Multiple weak classifiers are integrated for the final prediction. Yao et al. [34] proposed the MultiSource-TrAdaBoost algorithm to transfer knowledge from multiple sources and identify a source domain that is most relevant to the target domain among these source domains. The second strategy is to combine DDA to build subnetworks specific to multiple domains and finally take a weighted average of the prediction results of different subnetworks [37]. Zhu et al. [29] achieved a two-stage alignment by constructing shared subnetworks and non-shared subnetworks specific to multiple domains: feature space alignment for each pair of source–target domains and output alignment for each domain classifier. It should be noted that there may be domain shift between source domains, and it is inappropriate to directly merge multiple source domains into a single source domain.

3. Research methodology

3.1. Research framework

This study was carried out in two major steps, as shown in Fig. 1.

- Step 1: Single-source transfer learning model construction and source domain selection. The missing values of the original data were filled, and after the original data were normalized, the DTW algorithm was used to calculate the data distribution distance between the source domain and the target domain. Next, a single-source transfer learning model was built, Bayesian optimization was used to achieve hyperparameter optimization, and multiple sets of experiments were performed on each source domain in the source domain set. Finally, the applicability of DTW (i.e., whether DTW is suitable to measure the similarity between building energy consumption distributions) was verified according to the prediction accuracy of each experiment. The validity verification analysis of DTW is usually not omitted. Otherwise, the similarity measurement method can only be used as a reference in the selection of source domain, the results are not necessarily accurate [38–40]. This paper verifies the strong linear correlation between DTW and the prediction accuracy. Therefore, top four buildings with the smallest DTW values were selected as the source buildings for the multi-source transfer learning model.
- Step 2: Training and validation of multi-source transfer learning models. Due to the complexity of the model structure and the limitation of laboratory computing power, two of the four source buildings (Step 1) were chosen as the source domain of the multi-source transfer learning model to build the multi-source transfer learning model. Bayesian optimization was used to confirm the hyperparameters of the source model. Finally, the prediction performance of the multi-source transfer learning energy prediction model was evaluated and compared with that of the single-source transfer learning energy prediction model to verify the validity of the multi-source model.

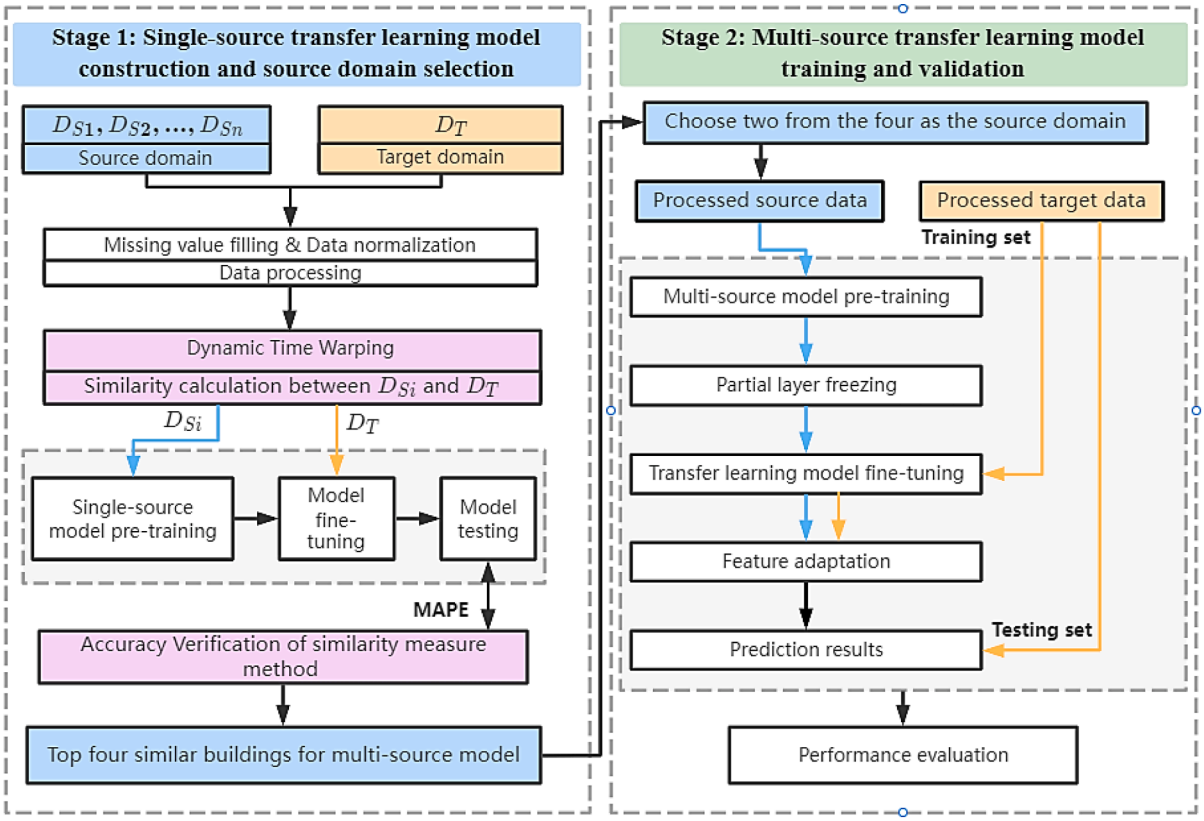


Fig. 1. Research outline.

3.2. Data pre-processing

In the field of deep learning, the quality of the data largely determines the quality of the model. Pre-processed data can enable the deep learning model to more accurately identify the changes between the data and thus obtain more accurate prediction performance. In this study, the missing values of energy consumption data were first filled with the mean values of their nearby data, and then multiple input features were normalized. Normalization can not only ensure that different features are at the same quantitative level but also accelerate the convergence speed of the model. In this paper, the min–max method is used to normalize the characteristic variables.

Min-max is one of the most useful normalization methods in the field of deep learning. It is suitable for situations in which the data do not conform to a normal distribution. The min–max method is shown in Equation (1):

$$\text{min - max} = \frac{x - X_{\min}}{X_{\max} - X_{\min}} \quad (1).$$

Where x is the value of the target feature and $X_{\max}$ and $X_{\min}$ are the minimum and maximum values of this feature, respectively. This equation limits the feature to between 0 and 1.

Due to the special structure of the LSTM used in this paper, the model itself cannot identify the input and output. The original data needs to be converted into a more model-friendly way so that the network can more accurately identify the time dependence of the original data. The sliding window method is widely used to construct the input sample set, which is only applicable to LSTM-related models [19]. Fig. 2 shows how this method works. The length of the original data is N, the length of the sliding window is Δt, and the prediction time range (i.e., the time step of each sliding window) is h. Note that the length of the sliding window needs to remain the same. According to the daily periodicity of building energy consumption, this paper sets the sliding window length Δt to 24; that is, every 24 h is used as a time series sample input into the model. The prediction time range h is set to 1; that is, the model is a single-step prediction, which predicts the energy consumption of the next hour. As the sliding window slides forward, the N samples of the original sequence are divided into $M = N\text{-}(\Delta t + h) + 1$ samples [41].

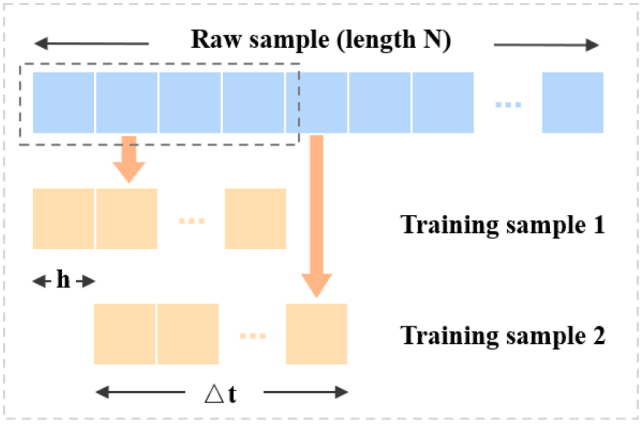


Fig. 2. Generation of training sample set.

3.3. Similarity measure between domains

The core problem of transfer learning is to find the similarity between the new problem and the original problem. The more similar the source domain and the target domain are, the better the knowledge in the source domain will be transferred to the target domain without causing negative transfer. In essence, the aim is to find a transformation that minimizes the distance (maximum similarity) between the source and target domains. There are various measures of similarity between two data domains in transfer learning, including spearman correlation

coefficient (SCC), JS divergence, DTW and MMD. DTW calculates the distance between two arrays. The calculation process is discrete and non-differentiable. Therefore, DTW cannot be used as a loss function of neural network and is more used as the basis of prediction task [42]. This study uses DTW to calculate the distribution distance between the source building energy consumption data and the target building energy consumption data so that the similarity between the source building energy consumption and the target building energy consumption can be quantified. At the same time, in order to select the distance calculation method that is most suitable for the energy consumption data set in this article, the JS divergence and SCC are also used to calculate the distance between source buildings and target buildings to compare with DTW. In contrast to DTW, MMD was first proposed for two-sample test to determine whether two distributions p and q are the same [43]. MMD is now widely used in transfer learning, especially as a part of the loss function in DDA. Therefore, MMD is used to compute the distance between the extracted latent features of the source and target domains.

3.3.1. DTW

DTW uses dynamic programming (DP) to calculate the distance between two time series, and it is widely used in isolated word speech recognition, information retrieval, and so on. It solves the template matching problem of different pronunciation lengths of the same word due to different speech speeds. In the field of load forecasting, DTW may capture excursions or extended periods of energy consumption peaks or troughs [27]. The DTW calculation process is as follows: Suppose there are two time series $X = \{(x_i)\}_{i=1}^{m}$ and $Y = \{(y_j)\}_{j=1}^{n}$. The DTW algorithm first constructs an $m \times n$ matrix network. The elements of the matrix (i, j) represent the distance $d(x_i, y_j)$ between x_i and y_j. The Euclidean distance is generally used (i.e., $d(x_i, y_j) = (x_i, y_j)^2$). The DP algorithm finds a path that passes through several lattice points in this matrix network, which is defined as a regular path and is represented by W:

$$W = w_1, w_2, \ldots, w_k \ \max(m,n) \leqslant k \leqslant m+n-1 \tag{2}$$

where $w_k = (i,j)_k$. The regular path W needs to satisfy the following constraints: the starting point of the path is (1, 1) and the end point is (m, n); if the path passes through point (i, j) in the network, then the previous lattice points that can reach this lattice point can only be $(i\text{-}1, j)$, $(i\text{-}1, j\text{-}1)$, and $(i, j\text{-}1)$. The purpose of the DP algorithm is to find an optimal path function when the above constraints are met (i.e., the mapping function of the time series X to Y) so that the cumulative distance D along the path W reaches the minimum value. The smaller the cumulative distance D, the higher the similarity between the two sequences. The optimal path and shortest distance DTW (X, Y) can be described as the following DP problem:

$$\begin{cases} \mathrm{DTW}(X,Y) = \min\{D(m,n)\} \\ D(i,j) = d(i,j) + \min\{d(i-1,j-1), d(i-1,j), d(i,j-1)\} \\ D(1,1) = d(1,1) \end{cases} \tag{3}$$

where $D(i,j)$ represents the local cumulative distance, the distance from (1, 1) to (i, j), and $D(1,1)$ represents the initial distance.

3.3.2. MMD

As the most frequently used similarity measure in transfer learning, the MMD calculates the distance between two distributions in reproducing kernel Hilbert space (RKHS), which is a kernel learning method [17]. Given a source domain $X_S = \{(x_S^i)\}_{i=1}^{N_S}$ and a target domain $X_T = \{(x_T^j)\}_{j=1}^{N_T}$, the MMD of the source and target domains in the RKHS can be defined as follows:

$$MMD(X_S, X_T) = \left\| \frac{1}{N_S}\sum_{i=1}^{N_S} \varphi(x_S^i) - \frac{1}{N_T}\sum_{j=1}^{N_T} \varphi(x_T^j) \right\|_H^2 \tag{4}$$

where φ is the nonlinear function in RKHS that is used to map the original variable into RKHS. $\mathscr{H}$ (RKHS) is a high-dimensional space with regeneration, and it has the property of inner product completeness for functions. Briefly, the MMD quantifies the distribution difference by calculating the distance between the source domain mean and the target domain mean in the RKHS, and the MMD distance occurs when the mean difference is greatest. The smaller the MMD value, the more similar the two domains are.

Considering that the data need to be mapped into a high-dimensional space, a Gaussian kernel is usually chosen as the kernel function [17]. However, MMD can only select one kernel function, and in most cases, we do not know which kernel is better at representation. To avoid the difficulty of selecting the kernel function in a single MMD and improve the representation ability of the model, this study selects a multi-kernel MMD (MK-MMD). MK-MMD uses the weighted average of multiple kernel functions to construct the total kernel function. For details on the MK-MMD, see [44].

3.4. LSTM

The LSTM network was proposed by Hochreiter et al. [45]. As a variant of the recurrent neural network (RNN), LSTM can capture the temporal correlation of sequences and learn long-term dependency information, and it is widely used in speech recognition, picture description, natural language processing, and so on. The structure of an LSTM unit (called a cell) is shown in Fig. 3. The LSTM cell is composed of three gate structures and cell states. Each unit has three input parameters: historical storage information C_{t-1}, current input information x_t, and current hidden information h_{t-1}. The cell state is equivalent to the information transmission path, allowing the information to be transmitted from C_{t-1} to C_t through several linear operation relationships, but the cell state cannot filter the information. The LSTM unit realizes the protection and screening of information through three gates equivalent to weights, which are the forget gate f_t, input gate i_t, and output gate o_t. The value range of the three gates is between 0 and 1, which is mainly realized by the operation of the sigmoid function and the dot product.

3.5. Multi-source transfer learning framework for energy prediction

3.5.1. Model structure

This study proposes a multi-source (two-source) transfer learning energy prediction model with LSTM as the feature extractor, which is based on parameter sharing and MK-MMD domain adaptation. The aim of using multiple sources is to improve the prediction accuracy of the model and make the model more general and robust. The structure of the multi-source model is shown in Fig. 4.

Given N source domains $\mathscr{D}_S = \{(\mathscr{D}_{S_i})\}_{i=1}^{N}$ and a target domain $\mathscr{D}_T$, the lstm1 and lstm2 are used to extract the time features of the input

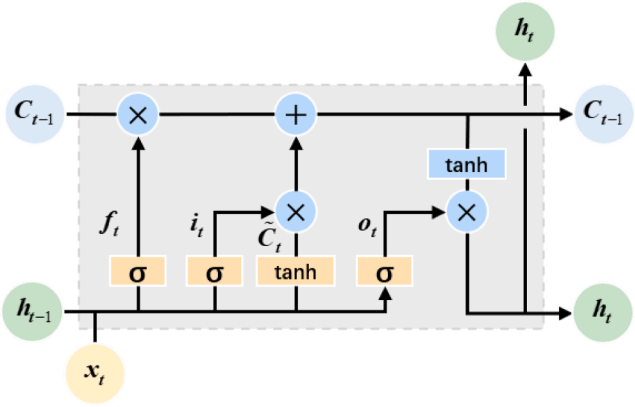


Fig. 3. Structure of LSTM cell.

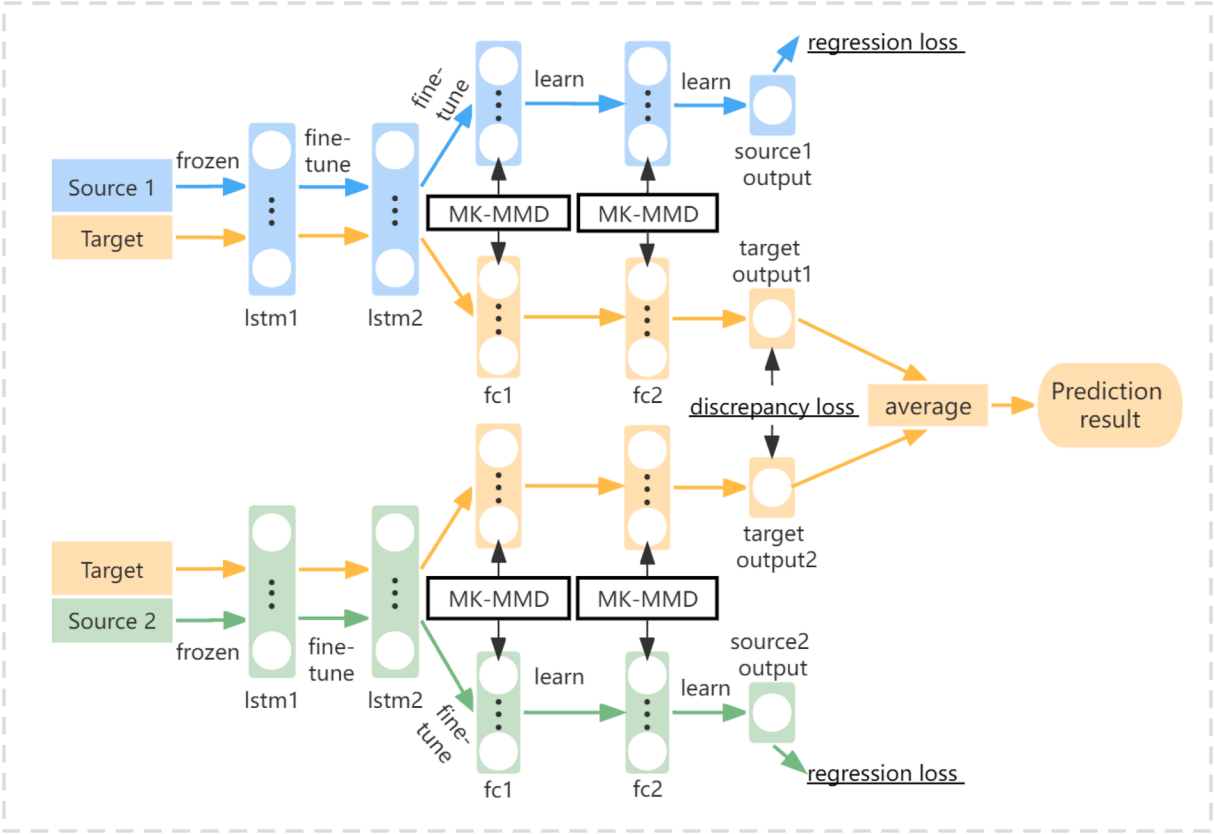


Fig. 4. Multi-source transfer learning energy prediction model structure.

time series, and the extracted time series features are input to the deep domain adaptation layers fc1 and fc2 (both fc1 and fc2 are fully connected layers). The purpose of the DDA layers is to learn the common features (domain-invariant representations) of the source domain and the target domain. In this study, the discrepancy-based domain adaptation method is selected, and the MK-MMD metric is added to the domain adaptation layer to reduce the distribution difference of source–target domain pairs by optimizing the distance loss function. It should be noted that when building the model, it is difficult to completely eliminate the shift between one source domain and one target domain, and when attempting to align multiple source and target domains, a greater mismatch may lead to unsatisfactory performance [29]. Therefore, inspired by [29], this paper builds separate subnetworks (the blue and green parts in Fig. 4) for multiple source–target domain pairs and extracts domain-invariant representations of the different source–target domain pairs. The idea of ensemble learning is used in model construction to construct subnetworks by training subnetwork parameters in two different source domains. Commonly used ensemble learning methods mainly include voting and averaging [46]. At a given prediction moment, the voting method assigns a higher weight to the intervals in which most of the predicted results of the base models are located. The voting method applies to classification.The averaging method includes arithmetic mean method and weighted mean method. Considering that the weight distribution is related to the multi-source transfer learning model itself, it is also related to the correlation between source data and target data and the similarity of different source data. Therefore, the weight distribution is a complex problem. As a simplification, arithmetic average method is chosen in this paper to take the mean value of the output values of the two subnetworks as the final prediction result.

3.5.2. Two stages of multi-source model training process

The model training process is mainly divided into two stages.

Pre-training stage Two source domains, $\mathscr{D}_{S_1}$ and $\mathscr{D}_{S_2}$, which are selected from $\mathscr{D}_S = \{(\mathscr{D}_{S_i})\}_{i=1}^{N}$, are input to the source model. The source model has no target domain data (no orange part in Fig. 4). The DDA layer becomes a simple linear layer (no MK-MMD). The source model is pre-trained with a large amount of source domain data, and the trained parameters are transferred to the target model.

Fine-tuning and relearning stage As shown in Fig. 4, multiple source–target domain pairs (It should be noted that in order to calculate the MMD later, the source domain intercepts the same time scale as the target domain) are individually input to the target model, the parameters of the lstm2 layer in the source model are fine-tuned, and multiple separate LSTM subnetworks extract the temporal features of the source–target domains individually. The extracted features are input into the domain adaptation layers fc1 and fc2 for model relearning. The MMD calculates the distance between the distribution of the source domain and the distribution of the target domain and optimizes this distance as part of the loss function. The optimization result enables the model to extract the common feature representation of the source and target domains. The output of the two subnetworks is averaged to get the final prediction result.

3.5.3. Loss function of model

The model has three goals: (1) to minimize the final regression loss, (2) to align the distribution of source–target domain pairs to minimize the loss of inter-domain discrepancy, and (3) to align the target predicted values of multiple subnetworks to minimize the discrepancy between the multiple predicted values. Therefore, as shown in Fig. 4, the loss function consists of three parts: regression loss, mmd loss, and discrepancy loss. The regression loss $\mathscr{L}_{\text{pred}}$ is defined as follows:

$$\mathscr{L}_{\text{pred}} = \sum_{i=1}^{N}\left(\frac{1}{N_S}\sum_{j=1}^{N_S} J_i(P_i(F_i(x_j^{S_i})), y_j^{S_i})\right) \tag{5}$$

where $J(\cdot,\cdot)$ is the mean square error (MSE) loss function,$P(\cdot,\cdot)$ is the regression predictor that obtains the predicted value of the model (i.e., the output layer in Fig. 4),$F(\cdot,\cdot)$ is a feature extractor that extracts common features of source–target domain pairs (i.e., the adaptive layer in Fig. 4), and N_S is the number of source domain data.

The mmd loss $\mathscr{L}_{\mathrm{mmd}}$ is defined as an estimate of the MMD:

$$\mathscr{L}_{\mathrm{mmd}} = \frac{1}{\mathrm{N}}\sum_{\mathrm{i}=1}^{\mathrm{N}} \mathrm{MMD}\left(\mathrm{F_i}(\mathrm{x}^{\mathrm{S_i}}), \mathrm{F_i}(\mathrm{x}^{\mathrm{T}})\right) \tag{6}$$

where $\mathrm{MMD}(\cdot,\cdot)$ is shown in Equation (4).

In the actual training process, the two subnetworks are trained from two different source domains, so the output predicted values may be different. However, in fact, a model should have only one prediction result for a given target value, so we take the absolute value of the difference between the target prediction values output by the two subnetworks as the discrepancy loss$\mathscr{L}_{\mathrm{disc}}$:

$$\mathscr{L}_{\mathrm{disc}} = \frac{1}{\mathrm{N_T}}\sum_{\mathrm{k}=1}^{\mathrm{N_T}} \left(\left|\mathrm{J_1}(\mathrm{P_1}(\mathrm{F_1}(\mathrm{x_k})) - \mathrm{J_2}(\mathrm{P_2}(\mathrm{F_2}(\mathrm{x_k}))\right|\right) \tag{7}$$

where N_T is the number of target domain data.

The total loss function formula of the model is as follows:

$$\mathscr{L}_{\mathrm{total}} = \mathscr{L}_{\mathrm{pred}} + \gamma(\mathscr{L}_{\mathrm{mmd}} + \mathscr{L}_{\mathrm{disc}}) \tag{8}$$

where$\gamma = \frac{2}{1+e^{-10\cdot\frac{i}{iteration}}} - 1,\ 0 \leqslant i \leqslant iteration$. To suppress the instability in the early stage of training, γ is gradually changed from 0 to 1.

3.6. Evaluation metrics

In this study, the coefficient of variation of the root mean square error (CV-RMSE) and MAPE were used to evaluate the predictive performance of the model. Due to the large differences in energy consumption levels among buildings, the use of the mean absolute error (MAE) and root mean square error (RMSE) is not appropriate [19]. The MAPE and CV-RMSE are calculated respectively as follows:

$$MAPE = \frac{1}{N}\sum_{i=1}^{N} \frac{|y_i - \widehat{y}_i|}{y_i} \tag{9}$$

$$CV - RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2}{N}} \Big/ \frac{\sum_{i=1}^{N} y_i}{N} \tag{10}$$

where $\widehat{y}_i$ and y_i are the predicted value and actual value, respectively, and N is the total sample size.

Detailed explanation of the algorithm proposed in this paper is summarized in Algorithm 1.

Algorithm 1: Multi-source transfer learning energy prediction network

Input: Source domain datasets selected by DTW: $\mathscr{D}_{\mathrm{S}} = \{(\mathscr{D}_{\mathrm{S_i}})\}_{\mathrm{i}=1}^{\mathrm{N}} = \{(\mathrm{X_{S_i}}, \mathrm{Y_{S_i}})\}_{\mathrm{i}=1}^{\mathrm{N}}$, target domain dataset: $\mathscr{D}_{\mathrm{T}} = (\mathrm{X^T}, \mathrm{Y^T})$

Output: parameters in trained network architecture

Pre-training stage:

1: Train a source model on source domain dataset and return the extracted features and parameters.

2: Pre-train a target model with parameters obtained from step 1.

Fine-tuning and relearning stage:

3: Give the number of training iterations T

4: For t in 1: T do

5: Randomly sample m of length 24 $\left\{x_j^{S_i}, y_j^{S_i}\right\}_{j=1}^{m}$ form one of N source domains $\{(\mathscr{D}_{\mathrm{S_i}})\}_{\mathrm{i}=1}^{\mathrm{N}}$

6: Randomly sample m $\left\{x_j^{T}\right\}_{j=1}^{m}$ from target domain dataset(X^T).

7: Feed source and target samples to lstm1 and lstm2 to get the time features $F_i(x_j^{S_i})$ and$F_i(x_j^T)$.

8: Feed $F_i(x_j^{S_i})$ and $F_i(x_j^T)$ to fc1 and fc2 to get the predicted values $P_i(F_i(x_j^{S_i}))$ and$P_i(F_i(x_j^T))$, the regression loss is computed as Equation (5)..

(continued on next column)

(continued)

Algorithm 1: Multi-source transfer learning energy prediction network

9: Use $F_i(x_j^{S_i})$ and $F_i(x_j^T)$ to calculate mmd loss as Equation (6).

10: Use $P_1(F_i(x_j^T))$ and $P_2(F_i(x_j^T))$ to compute the discrepancy loss as Equation (7).

11: Update the regression predictor P and the feature extractor F by minimizing the total loss in Equation (8).

12: End for

13: Return: The extracted features and parameters

4. Case study

This paper takes buildings in the Higashida area of Kitakyushu City, Fukuoka Prefecture, Japan as the experimental object. The dataset contains one-year power data and outdoor weather data of 40 buildings, such as office, commercial, and residential buildings, in the Higashida area. We use the historical data of the past 24 h to predict the building energy consumption in the next hour. First, we build a single-source transfer learning model and use DTW to calculate the distribution distance of each source domain and target domain. According to the correlation between the prediction accuracy of the single-source model and DTW, the rationality and accuracy of the DTW algorithm on the energy consumption data are verified. Then, we choose the four buildings most similar to the target building as the source domain set, build a multi-source (two-source) transfer learning model, and verify that the multi-source model we built has higher prediction accuracy than the single-source model. Finally, we verify that the multi-source transfer learning model has better versatility and stability by supplementary experiments.

4.1. Data setting

We randomly selected 21 buildings from the Higashida area building set: eight commercial buildings, five factory buildings, three office buildings, two gas stations, one museum, one hotel, and one railway building. Since these buildings were located in the same area, all buildings had the same meteorological conditions. We chose one railway building as the target building and the remaining 20 buildings as the source buildings. The dataset spanned one year, from April 1, 2013 to March 31, 2014, with a time granularity of 1 h, so 8760 samples per building could be used. In the transfer learning model, the time span of the target domain dataset was two months, from February 1, 2014 to March 31, 2014 (one month of data was used to train the model, and the remaining month was used for data validation and prediction). The time granularity was 1 h, so the target building had 744 samples.

The historical electricity consumption data were collected from the energy consumption monitoring platform to obtain the total electricity load of the building, including the electricity consumption of lighting, elevators, equipment and the energy consumption of the air conditioning system in summer. Fig. 5 shows the Gaussian kernel density estimates of energy consumption for the 21 buildings. The figure shows that the kernel density estimates for these buildings have roughly similar distributions, which satisfy the conditions for transfer learning. Meteorological features are from Fukuoka Weather Station data on the official website of the Japan Meteorological Agency [47].

In this paper, each dataset contains four numerical features (variables): outdoor temperature, outdoor wind speed, relative humidity, solar radiation intensity, which are closely related to the operation of building systems. Each dataset also contains three categorical features: hour type (0–23 h), day type (Monday–Sunday), and holiday type (whether it is a holiday or not), which are used as indicators of periodicity and occupancy and are digitized by one-hot encoding. The labels are hourly energy consumption data for each building. The model input consists of two parts. One is the four numerical features of the past 24 h, three classification features, and historical energy consumption. The other is the day type, hour type, and holiday type of the forecast day, which are considered known conditions on the forecast day.

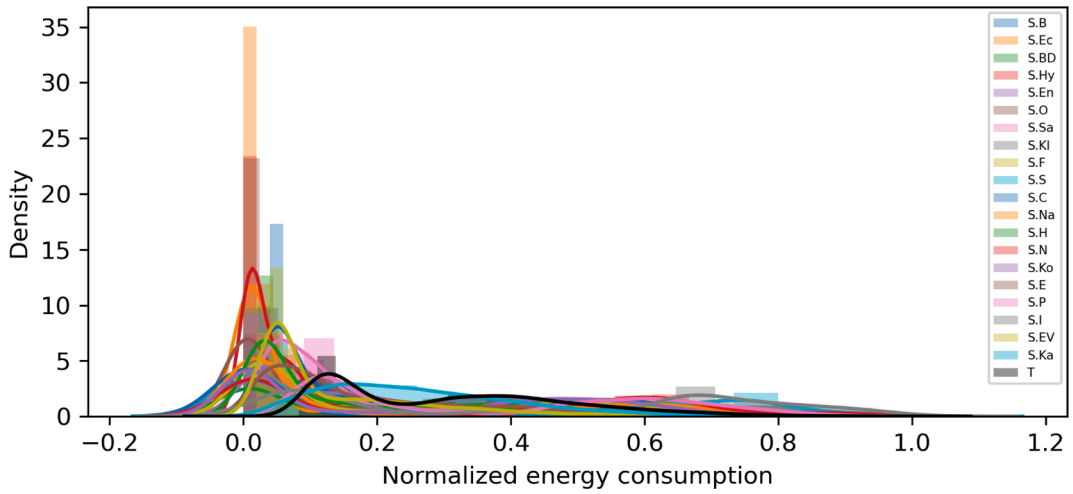


Fig. 5. Kernel density estimates of energy consumption per building (target buildings in black).

4.2. Experiments on single-source transfer learning energy prediction

This study first built a single-source transfer learning model. Compared with the model trained with only one month of target domain data, we expected the transfer learning model to have better predictive performance and the low historical load problem of the target building to be solved by transfer learning. In addition, we combined the DTW values to explore how similar the source and target domains are when the negative transfer phenomenon occurs.

First, the DTW algorithm was used to calculate the similarity of the normalized energy consumption data, and the distribution distances of 20 source domains and target domains were calculated individually. The DTW calculation results are shown in Table 1. We can see that the top three buildings that are most similar to the target railway building are an office, office, and factory. In addition, the DTW values of the same type of buildings, such as S.E and S.Ec (both offices), are not closer to that of the target railway building. Analyzing the reasons, the daily average load curves of S.E, S.Ec and target building T in each month (after normalization) are shown in Fig. 6. As can be seen from Fig. 6, the overall seasonal trends in energy consumption for the three buildings are similar. However, compared with S.E, S.Ec has a greater difference with T in the peak load of electricity consumption, and its energy consumption is obviously lower than that of S.E and T. From July to September and December to January, peak loads for S.E and T occur around 10am, while peak loads for S.Ec occur around 4 pm. Therefore, although S.E and S.Ec are of the same type, energy consumption may vary due to differences in occupancy and hours of operation.

Table 1
DTW calculation results.

Model ID	Building type	DTW	Model ID	Building type	DTW
S.E	Office	0.0619	S.BD	Commercial	0.1033
S.B	Office	0.0708	S.Ec	Office	0.1043
S.H	Factory	0.0761	S.EV	Gas station	0.1086
S.O	Factory	0.0821	S.Na	Factory	0.1128
S.Ka	Hotel	0.0835	S.Hy	Gas station	0.1183
S.Ko	Commercial	0.0943	S.N	Commercial	0.1277
S.En	Museum	0.0958	S.KI	Commercial	0.1429
S.F	Factory	0.0965	S.I	Commercial	0.1442
S.P	Commercial	0.0999	S.C	Commercial	0.1522
S.Sa	Hotel	0.1016	S.S	Commercial	0.1781

Note. It can be seen that the DTW of the same building type sometimes differ relatively obviously. On the contrary, sometimes the DTW of different building types are more similar.

4.2.1. Model setting

In this study, we set a total of 22 sets of models to evaluate the effectiveness of the transfer learning models. The task was to use the trained model to create rolling forecasts of the hourly energy consumption of the target building for two weeks from March 11 to March 24, 2014. The target domain and each source domain were combined to build a single-source transfer learning model, with a total of 20 experimental groups. In addition, two baseline models without a transfer learning framework, BaseM.1 (with only one month's data in the target domain) and BaseM.2 (with only one year's data in the target domain), were constructed to compare with the transfer learning model. Table 2 shows the division of source and target domains for the 22 experiments. The structure of the single-source transfer learning model was roughly the same as that of Fig. 4. The domain adaptation layer was a simple fully connected layer. The model structure is shown in Fig. 7. The structure of the benchmark model was the same as that of the transfer learning target model.

The single-source transfer learning model included two LSTM layers and two fully connected layers. On the validation set, a Bayesian optimization algorithm was used to optimize some hyperparameters, including the number of hidden layer units, input data batches, and learning rate. Due to the large number of models, it took too long to use Bayesian optimization for each model. In this study, only some model parameters were optimized according to the similarity between source domains, and other models were adapted one by one among the optimized parameters. The activation function was the ReLu activation function, and a batch normalization layer was added between the fully connected layer and the activation function layer. Batch normalization pulls the distribution of the input values of any neuron in each layer of the neural network back to the standard normal distribution. In this way, the value of the input activation layer falls in the region where the activation function is more sensitive to the input, which avoids the problem of gradient disappearance. The built-in Kaiming initialization of pytorch is used as the weight initialization method for the model, The Kaiming initialization improves the problem of mapping negative numbers to 0 in the ReLu layer in Xavier initialization, and the output value of each layer of the network is relatively stable [48]. In addition, to further improve the performance of LSTM, dropout of 0.2 was added to solve the overfitting problem, which is a process of randomly setting some model parameters to zero during model training. The epoch was set to 200. The parameter configuration of the neural network structure is shown in Table 3. The Adam optimizer was used to optimize the objective function during training, using python3.8.8 and pytorch1.8.1 + cuda10.2 to implement our model.

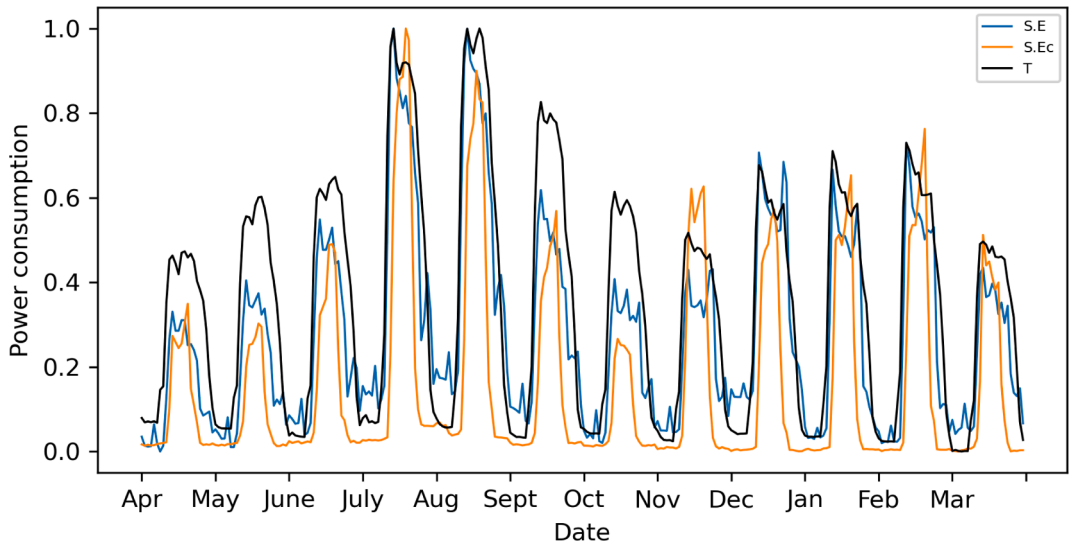


Fig. 6. Daily average curves of energy consumption for each month (S.E, S.Ec, T).

Table 2
Data divided into source and target domains.

Experiment	Hidden layer unit	Transfer learning framework	Source domain	Target domain
S.E, S.B, S.H, S.O, S.Ka, S.Ko, S.En, S.F, S.P, S.Sa, S.BD, S.Ec, S.EV, S.Na, S.Hy, S.N, S.KI, S.I, S.C, S.S	LSTM	Yes	Source building data from 2013.04 to 2014.02	Target building data from 2014.02 to 2014.03
BaseM.1	LSTM	No	No	Target building data from 2014.02 to 2014.03
BaseM.2	LSTM	No	No	Target building data from 2013.04 to 2014.03

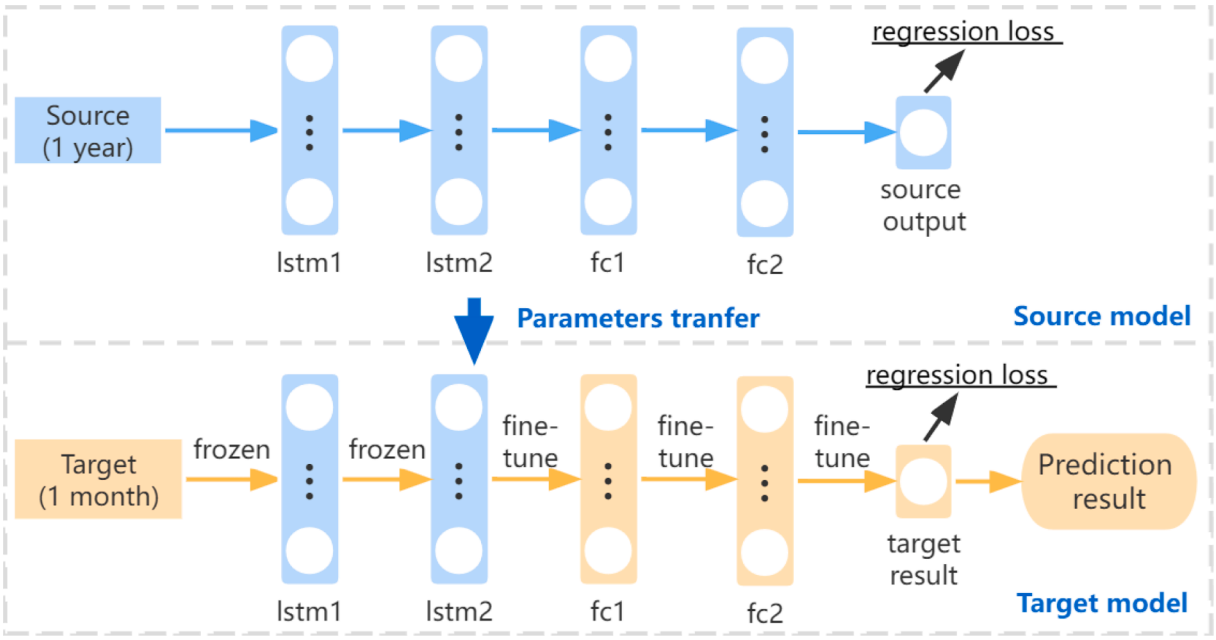


Fig. 7. Single-source transfer learning model framework.

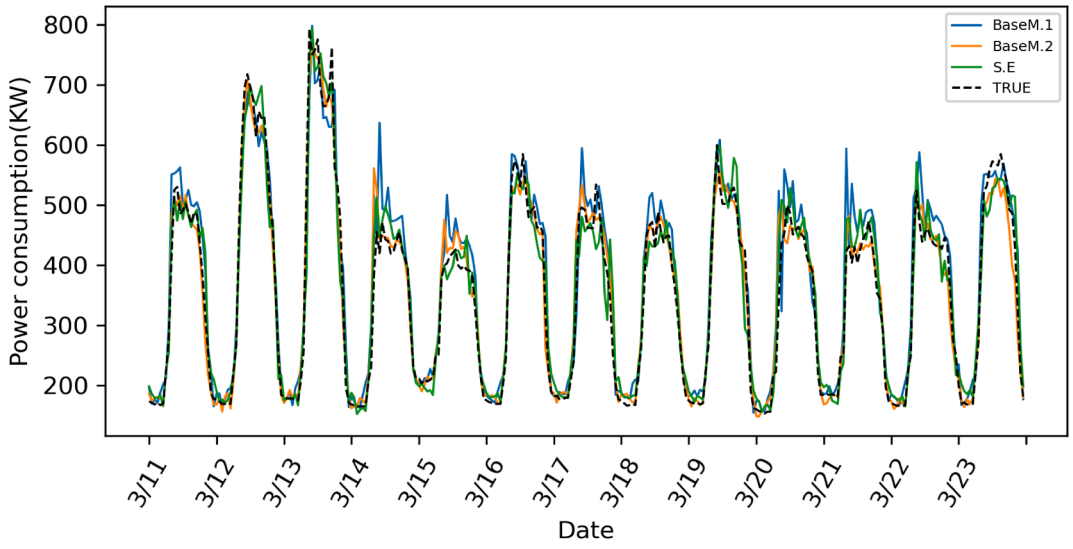
4.2.2. Experiment results and discussion

Fig. 8 shows the prediction results of some models, and the remaining single-source model prediction results are shown in the appendix Fig. A1. As can be seen from Fig. 8(a), the prediction result of BaseM.1 has the largest difference from the actual energy consumption, and the predicted values of BaseM.1 are mostly higher than the actual values at the peaks and troughs. This result is understandable, because only one month's data of the target building are used to train BaseM.1, and the training data are from February, but the real energy consumption in February is higher than that in March, which leads to high model prediction values. BaseM.2 shows the best predictive performance for actual energy consumption, which is reasonable, since we have enough

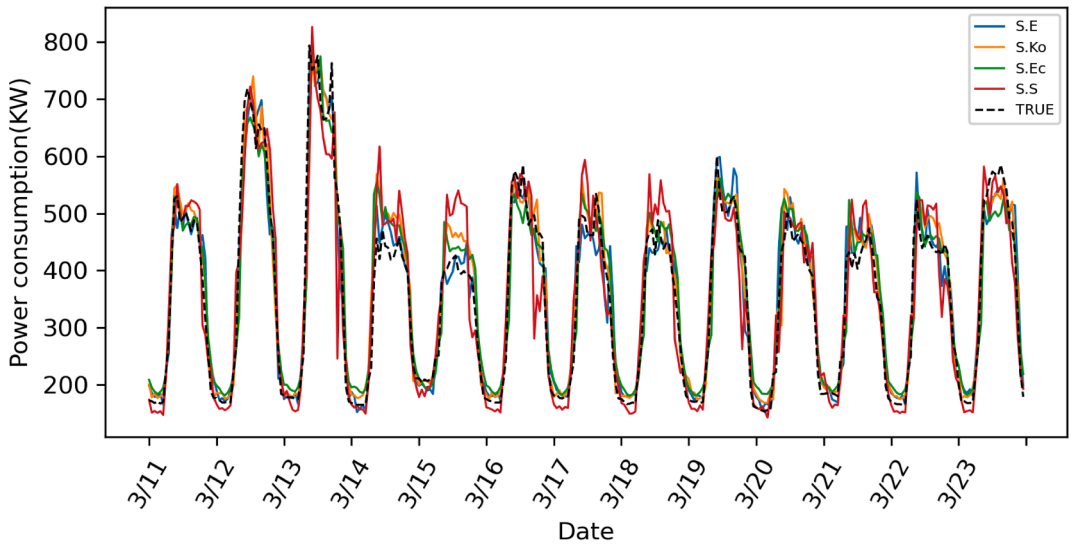
Table 3
Hyperparameter settings of different single-source transfer learning models and benchmark models.

Model ID	LSTM unit	Learning rate	Batch size	Model ID	LSTM unit	Learning rate	Batch size
S.E	(32, 32)	0.0049,lr*0.05	64, 24	S.Ec	(61, 61)	0.0104,lr*0.05	60, 14
S.B	(61, 61)	0.0104,lr*0.05	60, 18	S.EV	(140, 140)	0.0069,lr*0.05	45, 16
S.H	(32, 32)	0.0049,lr*0.05	64, 18	S.Na	(150, 150)	0.0080,lr*0.05	59, 16
S.O	(61, 61)	0.0104,lr*0.05	60, 16	S.Hy	(150, 150)	0.0080,lr*0.05	59, 16
S.Ka	(61, 61)	0.0104,lr*0.05	60, 16	S.N	(216, 216)	0.0070,lr*0.05	32, 16
S.Ko	(61, 61)	0.0104,lr*0.05	60, 16	S.KI	(216, 216)	0.0070,lr*0.05	32, 16
S.En	(32, 32)	0.0049,lr*0.05	64, 18	S.I	(216, 216)	0.0070,lr*0.05	32, 16
S.F	(32, 32)	0.0049,lr*0.05	64, 18	S.C	(32, 32)	0.0070,lr*0.05	64, 16
S.P	(150, 150)	0.0080,lr*0.05	59, 16	S.S	(216, 216)	0.0070,lr*0.05	32, 16
S.Sa	(61, 61)	0.0104,lr*0.05	60, 16	BaseM.1	(16,16)	0.1000	32
S.BD	(150, 150)	0.0080,lr*0.05	59, 16	BaseM.2	(110,110)	0.0001	16

Note. Here, the first value in the learning rate column is the learning rate of the source model. The second value is the learning rate of the target model during fine-tuning, and its value is 0.05 times the first value. The batch size is the same.



(a)



(b)

Fig. 8. Comparison of prediction results between different single-source transfer learning prediction models and benchmark models.

data for the target building for one year to train the deep learning model. The prediction results of S.E are between BaseM.1 and BaseM.2, which proves that using historical data of similar buildings to facilitate the energy prediction of target buildings can indeed improve the prediction accuracy. However, compared with BaseM.2, the prediction results of the source–target model show greater volatility.

As shown in Fig. 8(b), as the similarity to the target building gradually decreases, the prediction accuracy of the single-source transfer learning models gradually deteriorates overall. The prediction performance of some single-source transfer learning models with relatively high similarity is better than that of BaseM.1, which proves that transfer learning plays an important role in energy prediction and grid operation. The prediction performance of the single-source transfer learning models with low similarity is even worse than that of BaseM.1, indicating that the model exhibits negative transfer at this time. Table 4 presents the evaluation results of different models using the MAPE and CV-RMSE. We can see that the model built using the building most similar to the target building (i.e., S.E) achieves the best predictive performance.

To demonstrate the effect of transfer learning and the accuracy of DTW as a similarity measure more clearly, as shown in Table 4, the MAPE reduction rates of 20 single-source transfer learning models were calculated based on the MAPE value of BaseM.1. Then, we fitted a regression plot of the correlation between DTW values (see Table 1) and the rate of MAPE reduction with a confidence interval of 0.95, as shown in Fig. 9. It should be noted that we ignore individual value (eg S.Ka) that deviate from the overall trend, and observe the correlation between DTW values and MAPE reduction rates as a whole. Combining Table 4 and Fig. 9, we can clearly see that when the DTW value is greater than 0.11, the MAPE reduction rate of the single-source transfer learning models is less than 0, and the MAPE value is larger than that of BaseM.1. This indicates that when DTW $>$ 0.11, the data distributions of the source building and the target building are quite different, which will cause negative transfer due to the large domain shift. In addition, as the DTW value gradually decreases, the MAPE reduction rate gradually increases, and the two have an obvious negative correlation. This indicates that on our dataset, it is reasonable to use the DTW metric to measure the energy consumption similarity between the source building and the target building. In addition, this paper also uses JS divergence and SCC to calculate the similarity between the source building and the target building energy consumption data set. The regression analysis diagrams of JS divergence and SCC are shown in the appendix Fig. A2. It can be seen that JS divergence and SCC have no strong linear correlation with the final predictive accuracy of the model. Therefore, DTW is more suitable for this study as a similarity method.

4.3. Experiments on multi-source transfer learning energy prediction

In practice, we are likely to have multiple source domains, and the transfer learning model of a single source domain may be unable to learn all the features of the target domain. Thus, to verify the multi-source (two-source) transfer learning energy prediction model built in this study, the top four buildings with the smallest DTW values were selected as the source buildings (S.E, S.B, S.H, S.O), any two source buildings were combined into one source domain, and a total of six multi-source transfer learning models ($C_4^2 = 6$) were formed. The task was to use the trained model to predict the hourly energy consumption of the target building from March 11 to March 24, 2014. We expected that the multi-source transfer learning models would better extract the common features of the source and target domains and exhibit superior prediction performance compared to the single-source transfer learning models.

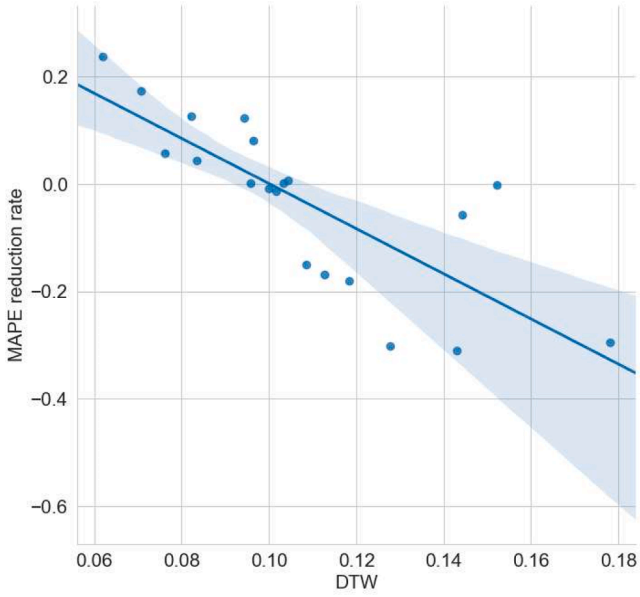


Fig. 9. Correlation fitting of DTW and MAPE reduction rate.

4.3.1. Model setting

We have already elaborated on the multi-source model structure (see Fig. 4) and training process in Section 3.5. Table 5 shows the settings of the training data, prediction data, source domain data, and target domain data of the multi-source model S.E + S.B, and the other five

Table 5
Data setup for model S.E + S.B.

Stage	Data source	Data content
Pre-training	Source data	Data on Buildings E and B (features and labels) from 2013.04 to 2014.01
Fine-tuning and relearning	Source data	Data on Buildings E and B (features and labels) from 2014.02 to 2014.03
	Target data	Data on the target building (features and labels) from 2014.02 to 2014.03
Prediction	Predictive data	Data on target building (features) from 2014.03.11 to 2014.03.24

Table 4
Prediction performance and MAPE reduction rates of different single-source and baseline models.

Model ID	MAPE	CV-RMSE	Reduction rate of MAPE (Compared to BaseM.1)	Model ID	MAPE	CV-RMSE	Reduction rate of MAPE (Compared to BaseM.1)
S.E	0.0739	0.1018	0.2381	S.Ec	0.0964	0.1217	0.0062
S.B	0.0801	0.1073	0.1742	S.EV	0.1115	0.1267	−0.1495
S.H	0.0847	0.1109	0.1268	S.Na	0.1134	0.1318	−0.1691
S.O	0.0914	0.1123	0.0577	S.Hy	0.1145	0.1371	−0.1804
S.Ka	0.0928	0.1189	0.0433	S.N	0.1262	0.1410	−0.3010
S.Ko	0.0850	0.1102	0.1237	S.KI	0.1270	0.1370	−0.3093
S.En	0.0968	0.1221	0.0021	S.I	0.1026	0.1160	−0.0577
S.F	0.0891	0.1148	0.0814	S.C	0.0971	0.1358	−0.0010
S.P	0.0978	0.1102	−0.0082	S.S	0.1256	0.1420	−0.2948
S.Sa	0.0983	0.1108	−0.0134	BaseM.1	0.0970	0.1267	0
S.BD	0.0968	0.1250	0.0021	BaseM.2	0.0637	0.0870	–

Table 6
Hyperparameter settings of different multi-source transfer learning models.

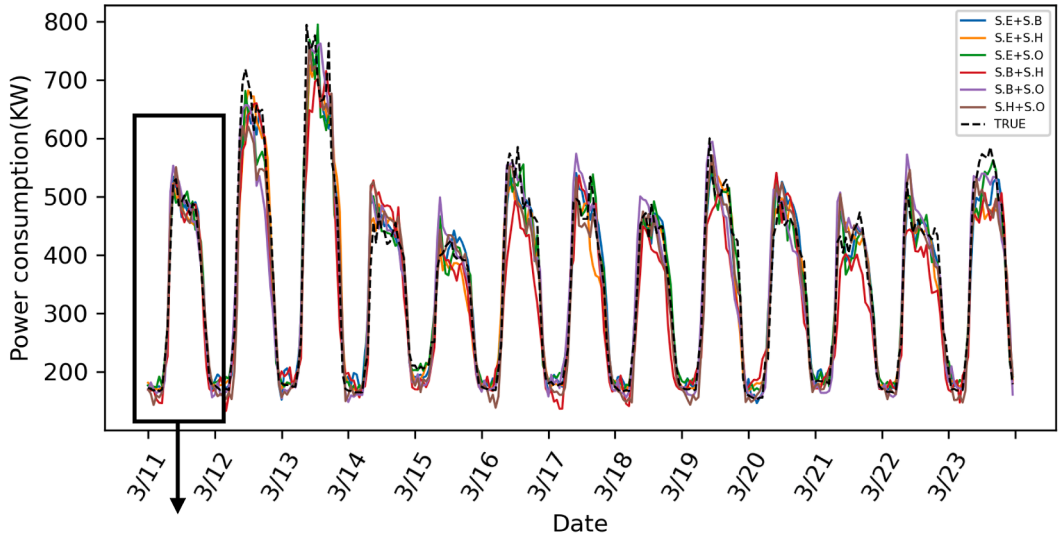
Model ID	LSTM unit	Learning rate	Batch size	Epoch
S.E + S.B	(167, 167)	0.0012, lr* 0.5, lr* 1	60, 20	200
S.E + S.H	(58, 58)	0.0030, lr* 0.5, lr* 1	43, 24	200
S.E + S.O	(234, 234)	0.0002, lr* 0.5, lr* 1	37, 16	200
S.B + S.H	(167, 167)	0.0012, lr* 0.5, lr* 0.5	60, 16	200
S.B + S.O	(127, 127)	0.0012, lr* 0.5, lr* 1	60, 20	200
S.H + S.O	(167, 167)	0.0012, lr* 0.5, lr* 0.5	60, 16	200

Note. In the learning rate column, the first value is the pre-training stage, the second value is the learning rate of some layers during fine-tuning, and the third value is the learning rate when the adaptive layer is relearning. lr is numerically equal to the first value. In the batch size column, the pre-training stage is the first value, and the fine-tuning and relearning stages are the second value.

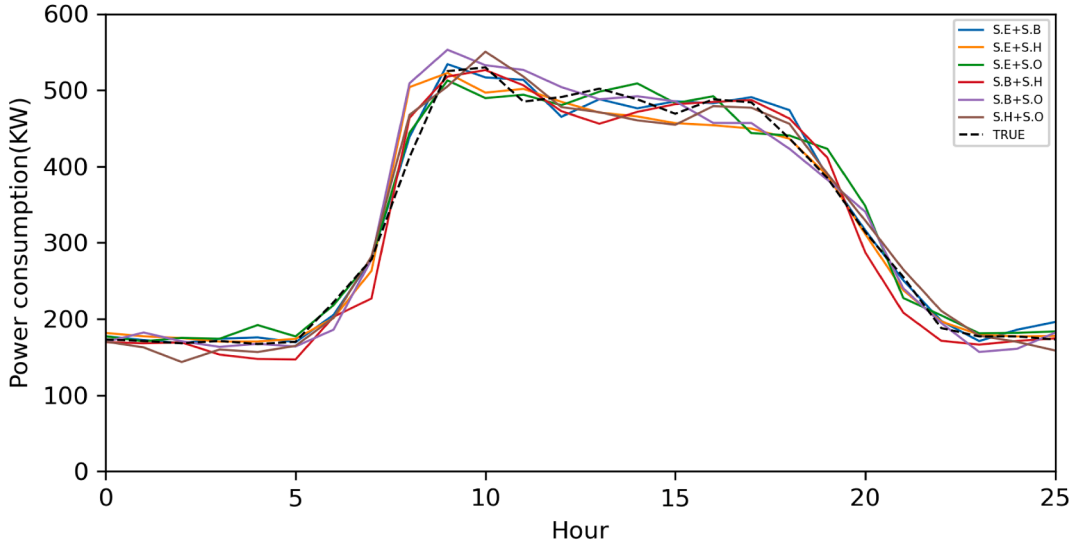
Table 7
Prediction performance of different multi-source transfer learning models and DTW across source domains.

Model ID	MAPE	CV-RMSE	DTW
S.E + S.B	0.0702	0.1013	0.0603
S.E + S.H	0.0712	0.1041	0.0554
S.E + S.O	0.0668	0.0941	0.0613
S.B + S.H	0.0795	0.1054	0.0722
S.B + S.O	0.0735	0.1095	0.0863
S.H + S.O	0.0796	0.1101	0.0601

models are consistent with S.E + S.B. As with the single-source models, due to the long training time of the Bayesian optimization algorithm, we used the Bayesian optimization algorithm on the validation set to optimize some hyperparameters of each multi-source source model,



(a)



(b)

Fig. 10. Comparison of prediction results of different multi-source transfer learning models.

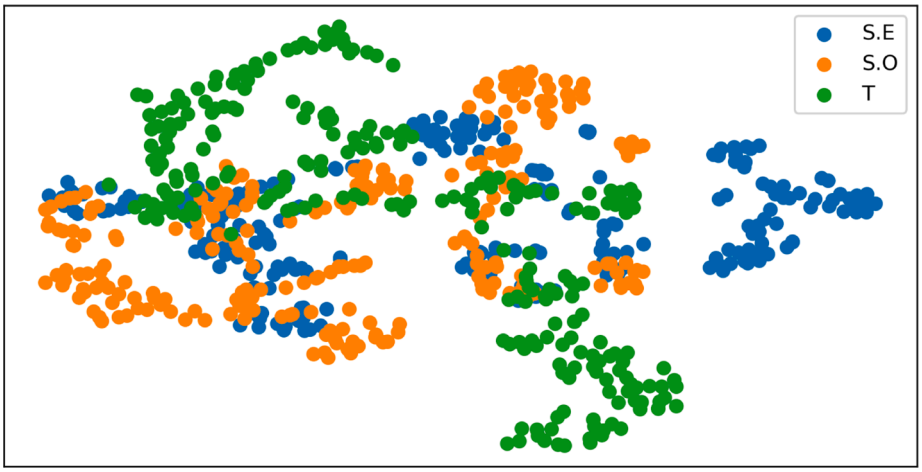
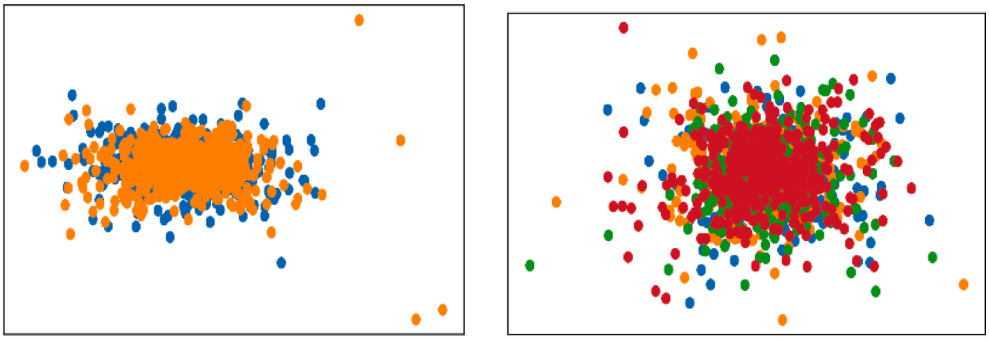


Fig. 11. Visualization of the original data features for the source and target domains.



(a) S.E → T. (b) S.E+S.O → T.

Fig. 12. Visualization of latent representations of source and target domains.

Table 8
DTW calculation results of target buildings and optimal source buildings.

Target building	S.E	S.O
T.M	0.0731	0.0806
T.Ni	0.0796	0.0922
T.Y	0.0580	0.0712

including the number of hidden layer units, input data batches, and learning rate. The number of Gaussian kernels was set to 5, the activation function adopted the ReLu activation function, the dropout value was set to 0.2, and the neural network parameter configuration of the multi-source transfer learning model is shown in Table 6.

4.3.2. Experiment results and discussion

Fig. 10(a) shows the energy prediction results of different multi-source models, and Fig. 10(b) presents the effect of magnifying the prediction results on March 11. A comparison with Fig. 8(b) reveals that the prediction results of the multi-source models are closer to the actual energy consumption. It is worth noting that the multi-source models significantly improved the results of the single-source transfer learning models whose prediction accuracy deteriorated due to the significantly lower peak load on March 14. Table 7 presents the results of evaluating six multi-source models using MAPE and CV-RMSE. It can be seen that for the multi-source transfer learning models, the CV-RMSE values of S.E + S.H and S.B + S.O are slightly higher than those of the single-source transfer learning models corresponding to the two source domains (see Table 4). Moreover, the prediction performance of the other four multi-source transfer learning models improved.

Table 9
Comparison of prediction performance on different target buildings.

Target building	Building type	MAPE		CV-RMSE	
		S.E	S.E + S.O	S.E	S.E + S.O
T.M	Office	0.1348	0.1340	0.1605	0.1587
T.Ni	Residential	0.2405	0.1856	0.2537	0.2100
T.Y	Hospital	0.0433	0.0429	0.0593	0.0550

For instance, S.E + S.O shows the best prediction performance: its MAPE value is 9.61 percentage points lower than that of the single-source model S.E and 21.13 percentage points lower than that of S.O, and its CV-RMSE value is 7.56 percentage points lower than that of S.E and 15.15 percentage points lower than that of S.O. Of the multi-source models, S.H + S.O exhibits the worst prediction performance, but compared with the corresponding single-source transfer learning models, its MAPE value is 12.91 percentage points lower than that of S.H and 6.02 percentage points lower than that of S.O. Its CV-RMSE value is 1.96 percentage points lower than that of S.H and 0.72 percentage points lower than that of S.O. This proves the superior performance of our multi-source transfer learning models compared to the single-source transfer learning models.

To explore the relationship between DTW and the multiple source domains of the multi-source model, we also calculated the DTW values between the two source domains of the multi-source models, as shown in Table 7. Interestingly, in the case where the two source domains are as similar as possible to the target domain, the larger the DTW value (i.e.,

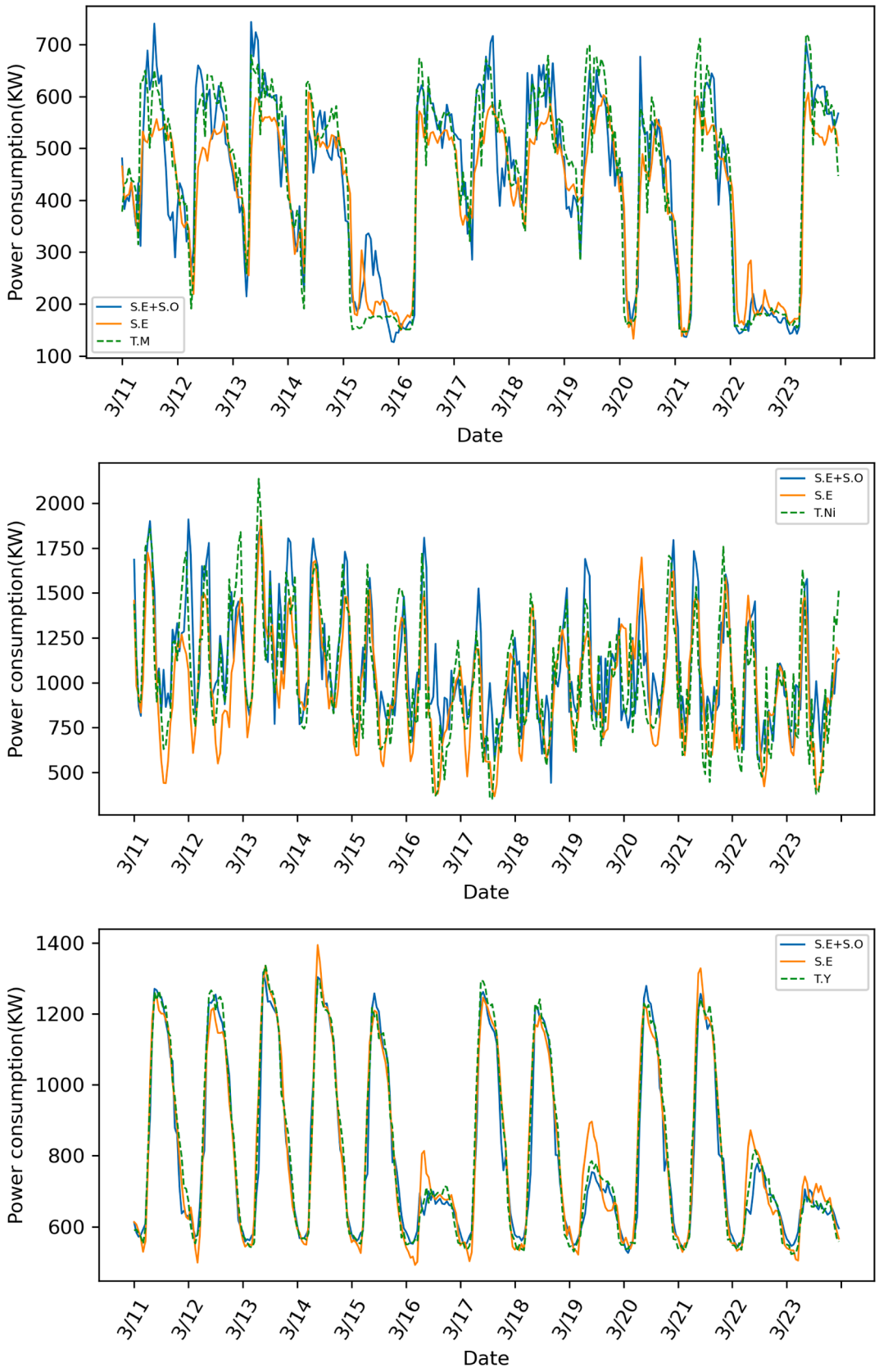


Fig. 13. Comparison of prediction results on different target buildings.

the greater the difference between the two source domains), the better the predictive performance of the model (as seen by comparing S.E + S.B, S.E + S.H, and S.E + S.O). A possible explanation is that as the difference between the two source domains increases, the contribution to the target domain may form a complementary relationship, and the multi-source model integrates this advantage to better characterize the underlying representations of features.

To further demonstrate the role of the MMD in extracting common

features from the source and target domains, Figs. 11-12 use t-SNE, a visualization tool that can reduce the dimension of high-dimensional data, to convert three-dimensional data to two-dimensional. Fig. 11 shows the t-SNE visualization of the original data features of the source and target domains (take S.E, S.O and T for example). We can see that there is a large domain shift in the original data features of different domains, which may be due to the different energy consumption habits caused by different building types. Fig. 12(a) shows the latent feature distribution extracted by the single-source transfer learning model (S.E → T, without the DDA layer). Fig. 12(b) show the feature distribution extracted by the two-source transfer learning model (S.E + S.O → T, with the DDA layer). Compared with Fig. 12(a), the distributions of the latent features in the source and target domains extracted in Fig. 12(b) are more uniform and concentrated, indicating that the two-source transfer learning model with the addition of the DDA layer can better extract the common features of the source and target domains.

4.4. Model adaptability analysis

To further verify the generality and stability of the constructed multi-source transfer learning model, we used the two models with the best prediction performance: the single-source transfer learning source model (S.E) in Section 4.3 and the multi-source transfer learning source model in Section 4.4 (S.E + S.O). The model structure was kept the same, but the target building was changed, and the two source models were fine-tuned on the target domain (learning rates were adjusted manually based on experience). The prediction performance of the single-source transfer learning models and the multi-source transfer learning models was then compared.

We randomly selected three buildings (office building, residential building, and hospital) as target buildings in the set of 40 buildings in the Higashida area, and these three buildings are not in the source building set in Section 4.2. The DTW values between the three target buildings and the source buildings S.E and S.O are shown in Table 8. The prediction performance of the model was also measured by MAPE and CV-RMSE metrics, and the results are shown in Table 9 and Fig. 13. It can be clearly seen that the MAPE and CV-RMSE of S.E + S.O on the three target buildings are both smaller than the corresponding values of S.E; S.E + S.O improves the prediction accuracy by an average of 8.11% in MAPE over the three sets of experiments. This shows that the multi-source transfer learning model is more general. Due to the different building types, the actual energy consumption curve fluctuates differently, but the overall MAPE of S.E + S.O is kept within 20%, and the CV-RMSE is generally kept within 21%, which indicates that the multi-source transfer learning model we constructed has good stability and can better adapt to the energy prediction of different building types.

5. Conclusions

This paper proposes a novel multi-source (two-source) transfer learning energy prediction model based on LSTM and MK-MMD domain adaptation for the short-term energy prediction of buildings lacking historical data. First, the model is trained on multiple source domain data with a time span of one year. Then, the trained source model parameters are transferred to the target model, and the target model is trained with one month's source and target domain data. The model integrated multiple subnetworks, and LSTM extracts the temporal features of each pair of source–target buildings separately. The distribution distance between source–target domain pairs is reduced by domain adaptation based on MK-MMD so that the model can finally extract the common features of source–target domain pairs. In addition, this paper uses DTW as the similarity measure to select the best source domain set. The prediction performance of the multi-source transfer learning model is evaluated through single-source and multi-source experiments. The main conclusions are as follows:

(1) Compared with the single-source transfer learning model, our proposed multi-source transfer learning energy prediction model has better prediction performance and can better adapt to the situation in which there are multiple different types of source buildings in practical applications. The MAPE of the multi-source model improves the prediction accuracy by 6.88–15.37%. Supplementary experiments prove that the multi-source model has better generality and stability and shows better prediction accuracy on target buildings of different building types. Compared to S.E, S.E + S.O improves the prediction accuracy by an average of 8.11% in MAPE.
(2) Plotting the correlation fitting graph of the MAPE reduction rate and DTW of the single-source model showed that DTW is significantly negatively correlated with the MAPE reduction rate, indicating that on the experimental dataset, DTW can measure the similarity between building energy consumption datasets relatively accurately.
(3) The source building data do not always improve the prediction performance of the target building, which often depends on the similarity between the source and target buildings when the model structure is the same. In this paper, a single-source transfer learning model with DTW < 0.11 can improve the prediction performance of target buildings compared to a model trained on only one month of target data.

Our future research will focus on the following aspects: (1) This paper does not consider the situation of more than two source domains, and we will consider expanding the source domains to three or even four in the future. We plan to explore in more detail the effect of the number of source domains on the prediction performance of our proposed multi-source model. (2) The time length of the source and target domains has a certain influence on the prediction performance of the model. This paper fixes the time length of the source and target domain data. In the future, it is necessary to study the influence of the source and target domain data length on transfer learning further. (3) In addition, this paper focuses on the comparison of single-source and multi-source models, so other machine learning models, such as SVR, GRU, and CNN, are not considered. Contrastive experiments with other neural networks should be conducted in the future.

CRediT authorship contribution statement

Huiming Lu: Formal ananlysis, Methodology, Software, Validation, Writing – original draft. **Jiazheng Wu:** Writing – review & editing, Conceptualization. **Yingjun Ruan:** Resources, Funding acquisition. **Fanyue Qian:** Project administration, Data curation. **Hua Meng:** Supervision. **Yuan Gao:** Visualization. **Tingting Xu:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix

(See Fig. A1, Fig. A2).

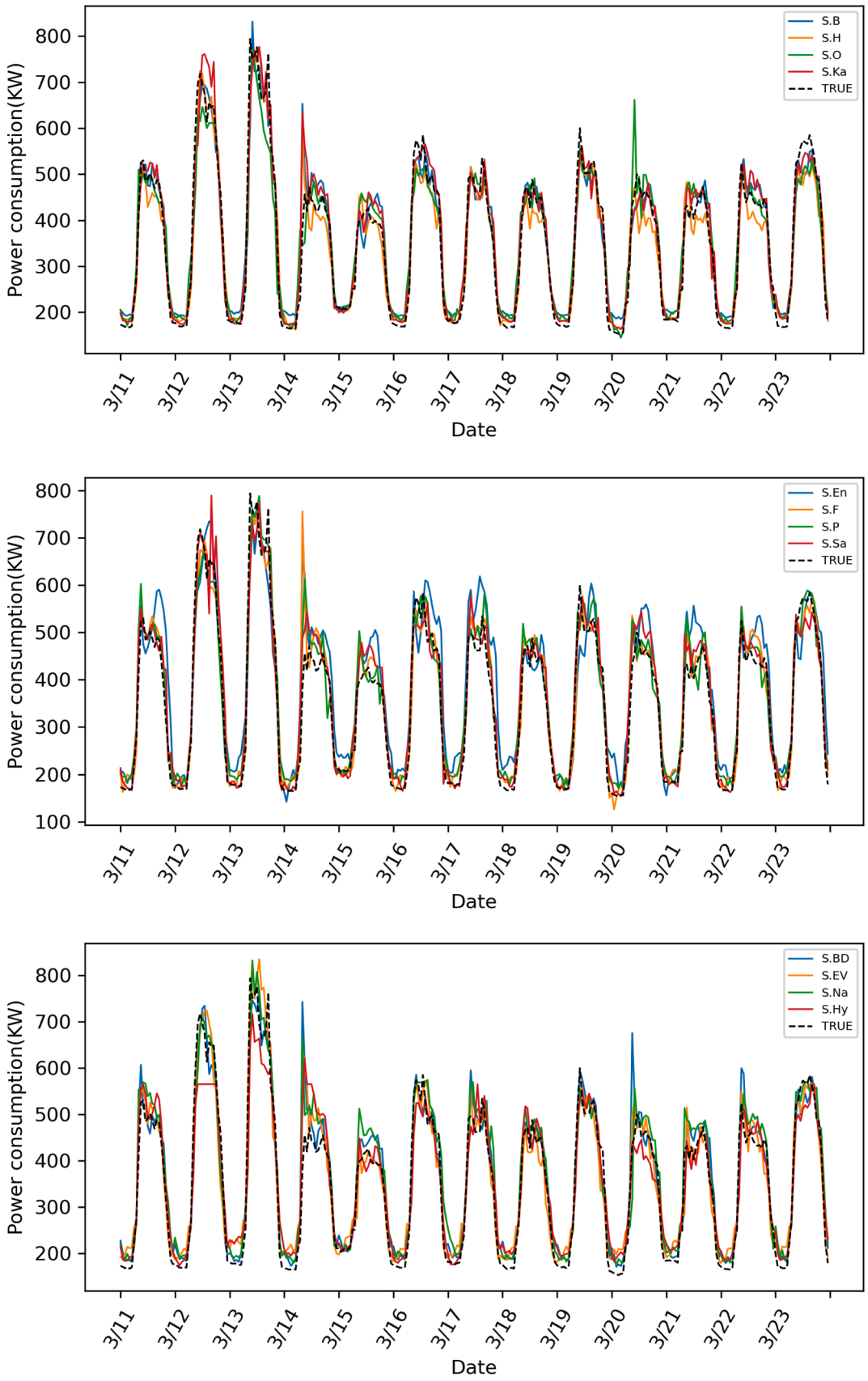


Fig. A1. Comparison of prediction results between different single-source transfer learning prediction models.

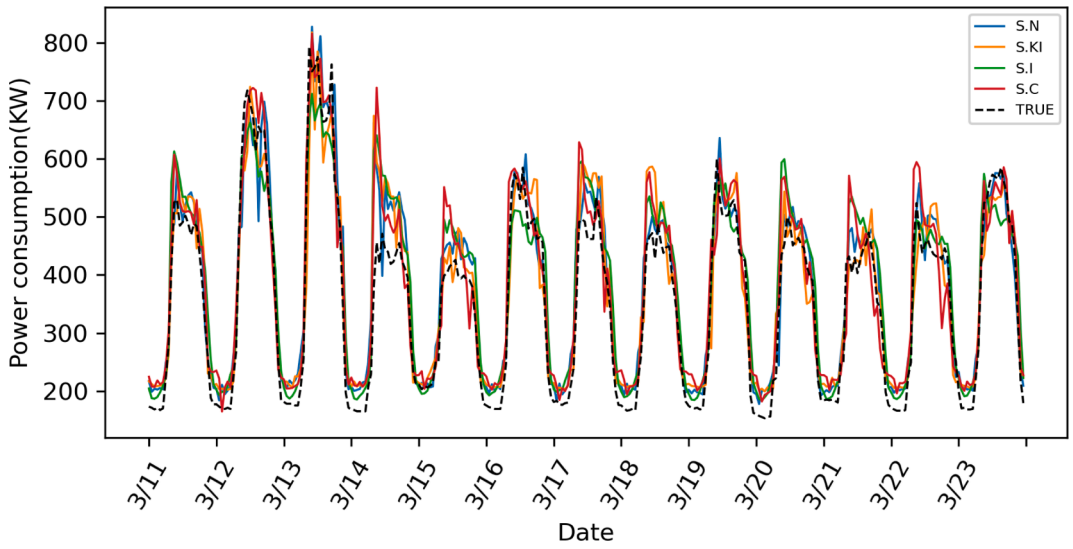


Fig. A1. (*continued*).

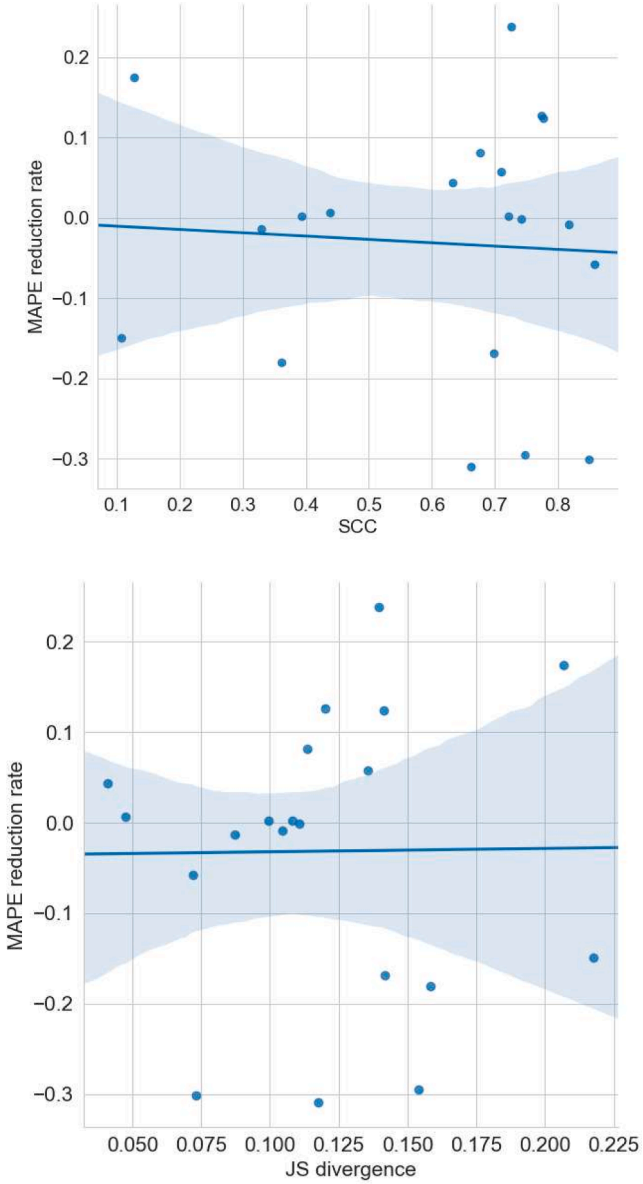


Fig. A2. Correlation fitting of SCC and JS divergence with MAPE reduction rate, respectively.

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